BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It lets the OS communicate with/control a specific hardware device.

Q2. A link/reference to the next node (for a singly linked list).

Q3. In its left subtree (under the chosen no-duplicate convention).

Q4. Its element locations can be calculated from base address and index.

You Do pages · Independent questions

Q1. Desktop publishing/page-layout software [2], control of typography/image placement [2], export/print layout advantage [2].

Q2. Applications solve user tasks [2]; utilities maintain/protect/optimise system [2]; relevant examples [2].

Q3. Database/inventory app [2], backup utility [2], accessibility/security [2], balanced cost/training/connectivity limitations [2].

Exit Ticket · Model answer

False. A word processor is an application.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

LESSON 112 · Lesson 112 · Choosing Applications and Utility Software

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What role does a device driver perform?

Q2. A linked-list node has data and what other essential field?

Q3. In a BST, where is a value less than the current node placed?

Q4. Why can an array provide direct indexed access?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

application software — Programs that meet a user task.

utility — Software for maintenance, protection or optimisation.

DBMS — Database management system.

backup — Recoverable copy to protect against data loss.

New learning

Clarify needs — Identify users, scale, accessibility, platform and constraints.

Match applications — Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

Understand utilities — Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Worked example

A clinic manages thousands of appointments accessed concurrently. Spreadsheet or database application?

1. Model answer — A DBMS is better: structured records, querying, concurrency, constraints and access rights. A spreadsheet is useful for small-scale analysis but is not ideal as a multi-user transactional store. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

A designer produces a precise engineered component and needs automated backups. Which software types?

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

CAD suits accurate technical modelling with dimensions and revision. A backup utility produces recoverable copies; design software and utility software have different purposes.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Justify an application for editing a school magazine with text and complex page layout. [6]

Question 2

Differentiate application software from utility software using two examples. [6]

Question 3

A small charity needs inventory records, backups and accessible user interfaces. Recommend software and justify limitations. [8]

Exit Ticket · Answer before checking

A backup utility and a word processor are both examples of utility software.